

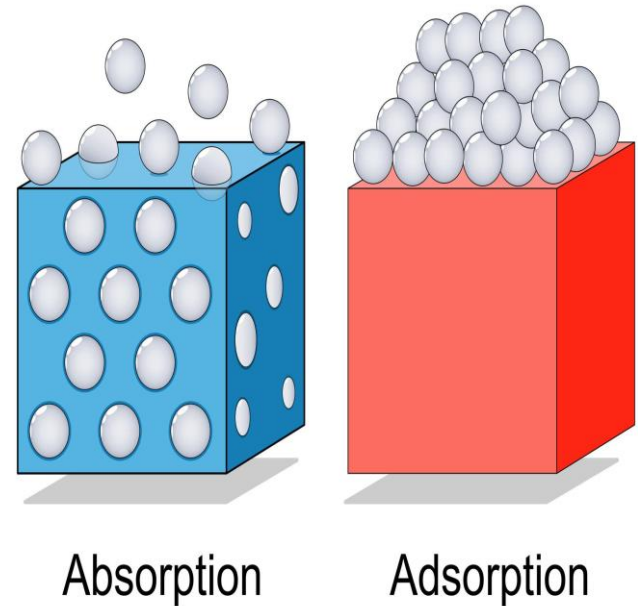
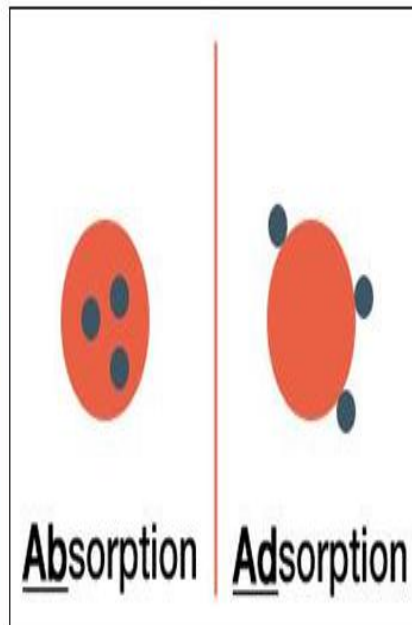
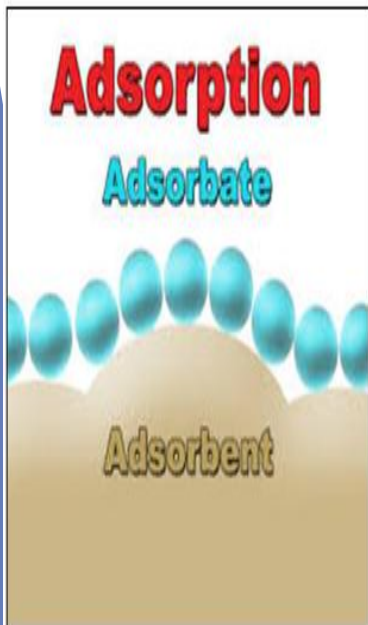
ADSORPTION AT INTERFACES

► **ADSORPTION:**

- It is surface effect.
- E.g. concentration of alkaloid molecule on the surface of clay.

► **ABSORPTION:**

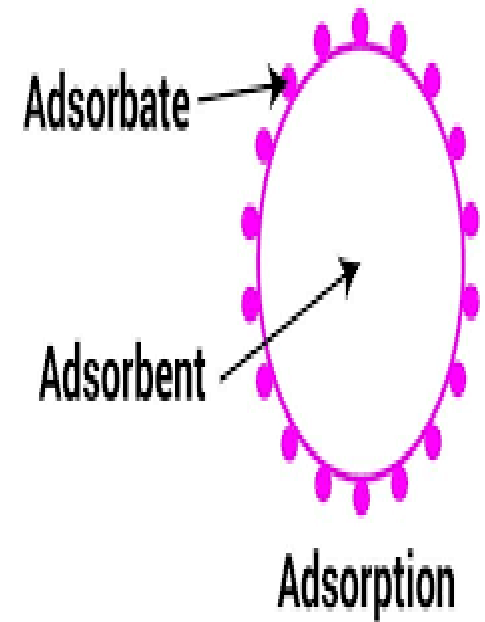
- Gas or liquid penetrates into the capillary spaces of absorbing medium.
- The taking up of water by a sponge is absorption.



Adsorption at solid interfaces

Adsorption is the adhesion of ions or molecules of gas, liquid, or dissolved solutes to a solid surface.

- ❑ the process of trapping or binding of molecules to the surface is called adsorption
- ▶ **Adsorbent**: it is the solid on its surface adsorption takes place.
- ▶ **Adsorbate**: the substance which is adsorbed on the surface of adsorbent



Adsorption at solid interfaces

- ▶ Similar to surface tension, adsorption is a consequence of surface energy. atoms on the surface of the adsorbent are not wholly surrounded by other adsorbent atoms and therefore can attract adsorbates.
- ▶ Adsorption reduces the imbalance of attractive forces which exists at a surface, and hence, the surface free energy

Adsorption at solid interfaces

- ▶ Adsorption at solid interfaces is classified according to the type of attractive force holding the molecule to the surface into:
 - a. Physical adsorption:** involves weak & reversible interaction between the solid and the adsorbate (van der Waals forces).
 - b. Chemical adsorption** (chemisorption): involves chemical & irreversible interaction where the adsorbed layer is combined by covalent forces to the solid.

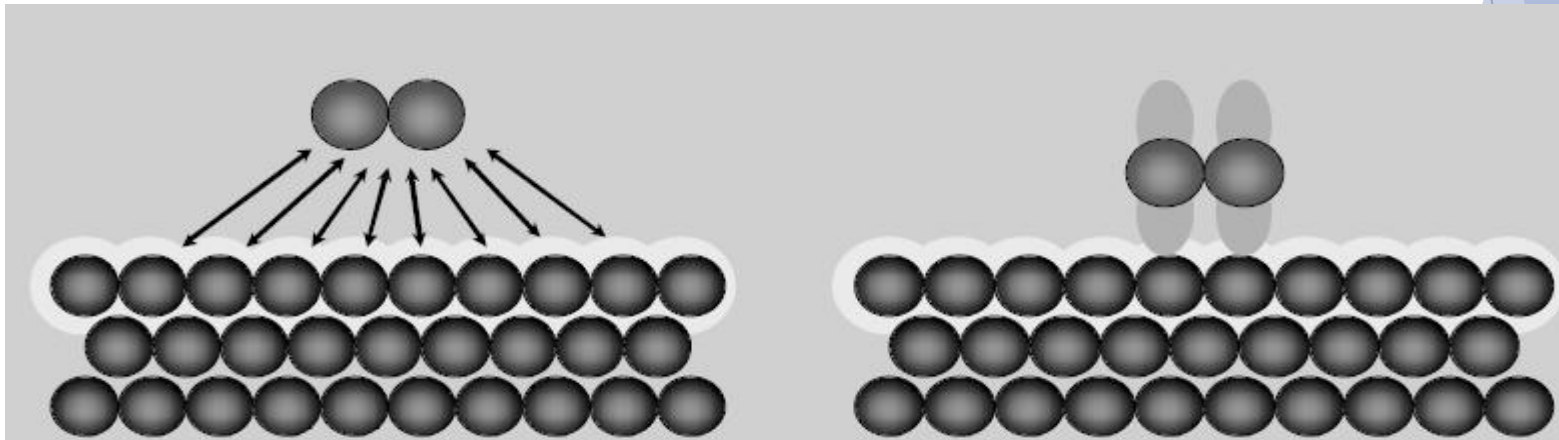
Adsorption at solid interfaces

➤ **Sorption:** when both adsorption and absorption happen together.

➤ **Desorption :**
adsorbed molecules or ions are removed from the solid surface.

Type of adsorption

Physical adsorption	Chemical adsorption
Reversible	Irreversible
Weak van der Waals forces	Strong chemical bond
Non-specific	More specific
e.g. adsorption of gases on charcoal	e.g. adsorption of oxygen on silver or gold.



Adsorption at solid/gas interface

► Adsorption of gas is important in following area

1. Removal of objectionable odors from the room.
2. Prevention of obnoxious gases entering into the body (gas masks)
3. Estimation of surface area and particle size of powders;

The amount of gas adsorbed per unit area or unit mass of solid is measured at different pressure of the gas at constant temp.

Adsorption at solid/gas interface

- ▶ The degree of adsorption of gas by a solid depends on
 1. Nature of adsorbent
 2. surface area of the adsorbent.
 3. Nature of adsorbate
 4. partial pressure of gas.
 5. Temperature.

Freundlich isotherm

- ▶ The relationship between pressure of the gas and amount adsorbed at constant temp has been expressed by freundlich isotherm equation:

$$\frac{x}{m} = kP^{1/n}$$

- ▶ Where x = wt of gas adsorbed per unit wt of adsorbent, m
- ▶ P= equilibrium pressure, k and n = constant.
- ▶ This equation gives linear graph when (x/m) is plotted against pressure p.
- ▶ The constant k (Freundlich constant (indicates adsorption capacity))

Freundlich isotherm

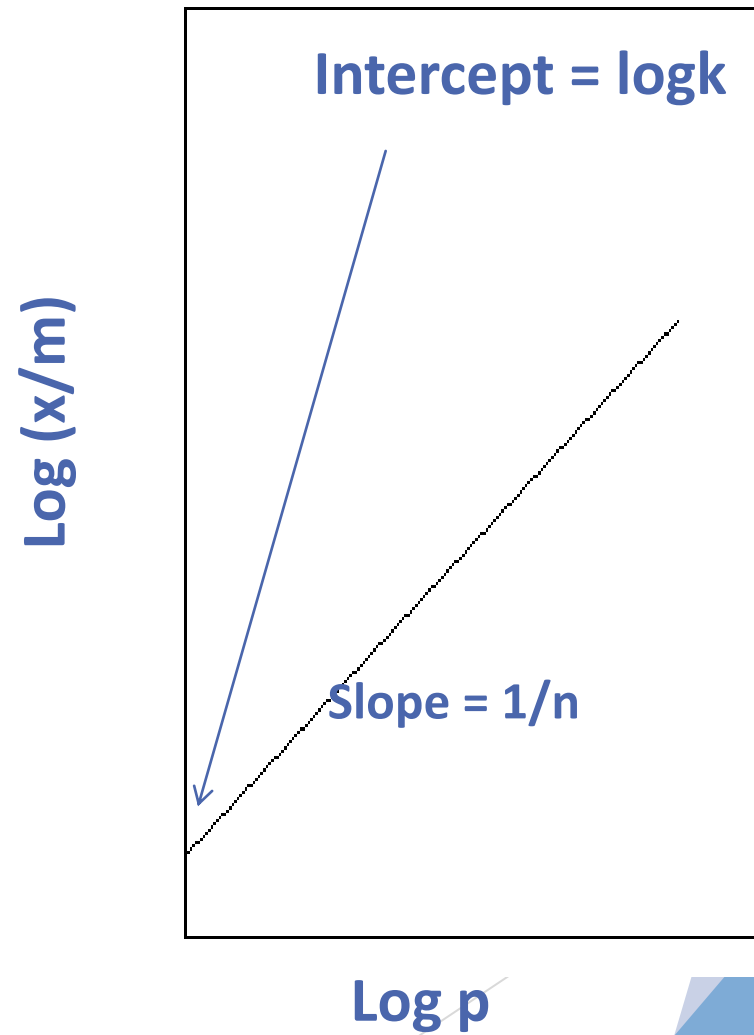
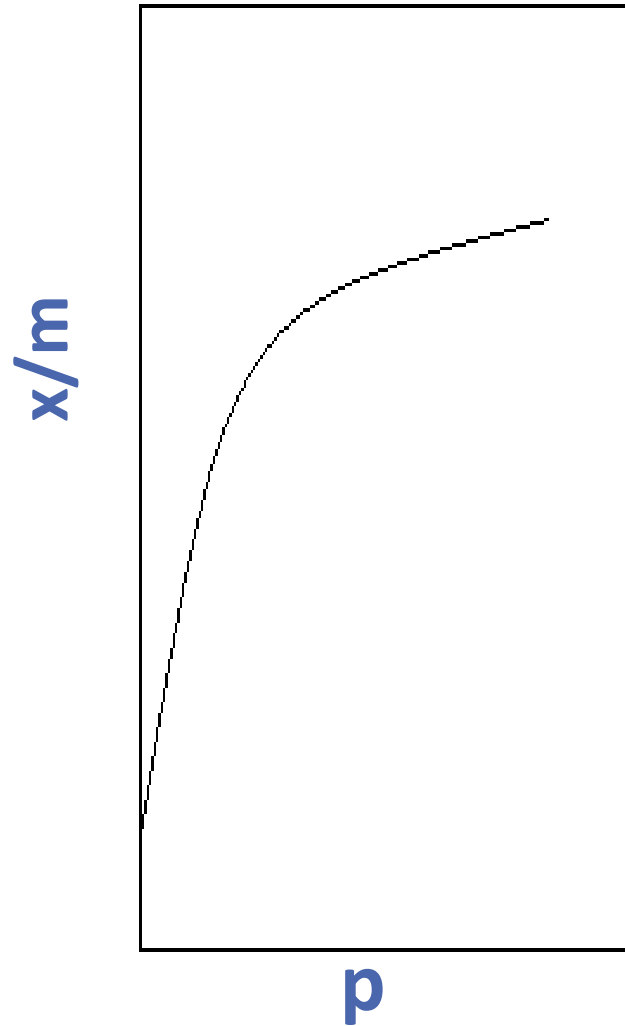
- ▶ n (Freundlich constant) tells you **how strong and favorable the adsorption is**. It is evaluated from the experiment and depends on temperature and nature of the adsorbent and adsorbate.

- ▶ Convert the eqn in logarithmic form

$$\log \frac{x}{m} = \log k + \frac{1}{n} \log P$$

- ▶ Thus, plotting $\log (x/m)$ versus $\log p$ gives a straight line with the slope $(1/n)$ and the intercept on the y axis is k .

Freundlich isotherm



ADSORPTION OF MOLECULES OF SOLUTE ON SOLID SURFACES

Factors affecting adsorption

1. Solute concentration:

- ▶ Increase solute conc increases amount adsorbed

2. Surface area of the adsorbent

- ▶ Reduction of particle size of adsorbent will increase surface area and then increase adsorption

3. Temperature

- ▶ Adsorption processes is usually exothermic. So increase temp will decrease adsorption amount

Factors affecting adsorption

4. Removal of adsorbed impurities

- ▶ Adsorbed impurities such as gases or moisture on the surface of adsorbents decrease their efficiency.
- ▶ Heating at 110°C for 1 hr will remove these impurities
- ▶ Such as gases or moisture on the surface of solids adsorbent $\rightarrow$ $\downarrow$ the efficacy of these adsorbents

5. Solvent competition

- ▶ Low affinity between solute and solvent (sparingly soluble) leads to higher adsorption
- ▶ Solutes are readily adsorbed when they are sparingly soluble in the solvent and when the solvent has a low affinity for the adsorbent.

6) Molecular weight

Higher molecular weight solutes are more readily adsorbed than low molecular weight solute this is due to Van Der Waals forces of attraction which increases with the size of molecules.

General applications of adsorption

▶ 1. Decolorization

- ▶ It is used by adding 1% charcoal with the substance solution, shaking and filtration to give a colourless solution
- ▶ Charcoal may adsorb organic and inorganic compounds
- ▶ e.g., clarification of aromatic water.



General applications of adsorption

► 2. Adsorption of water vapour (Drying agents)

Silica gel, aluminium oxide and clay are used as adsorbents. Alumina and silica are commonly used as they remain dry after adsorption

- It is common knowledge that small satchets containing silica gel are used in electronic equipments to keep them dry, by adsorbing water vapor, which is always present in the humid weather conditions.

General applications of adsorption (cont)

▶ 3. medicinal use

- ▶ Kaolin has the ability to adsorb both acidic and basic intestinal toxins.
- ▶ Activated charcoal is carbon exposed to elevated temperature to remove adsorbed gases. The treatment results in a highly porous charcoal. These tiny holes give the charcoal a surface area of 300-2,000 m²/g, allowing liquids or gases to pass through the charcoal and interact with the exposed carbon.

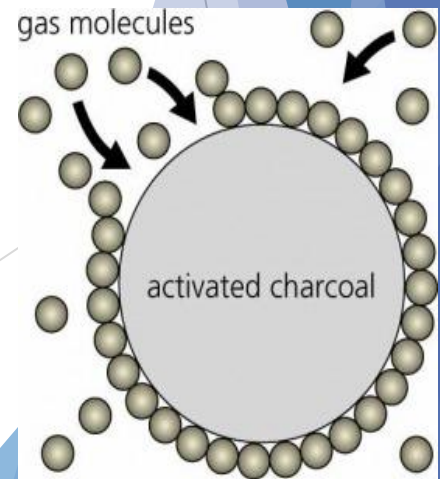
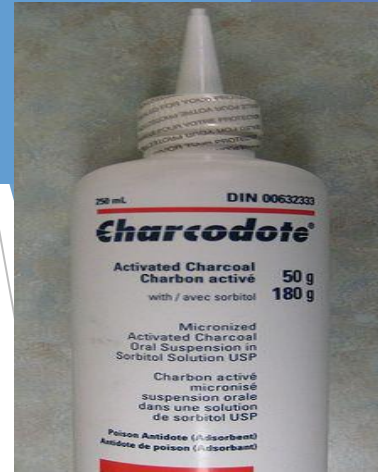
General applications of adsorption (cont)

▶ 4. medicinal use (cont.)

- ▶ The carbon adsorbs a wide range of impurities and contaminants, including chlorine, odours, and pigments.
- ▶ Because adsorption works by chemically binding the impurities to the carbon, the active sites in the charcoal eventually become filled.
- ▶ Activated charcoal filters become less effective with use and have to be recharged or replaced.

General applications of adsorption (cont)

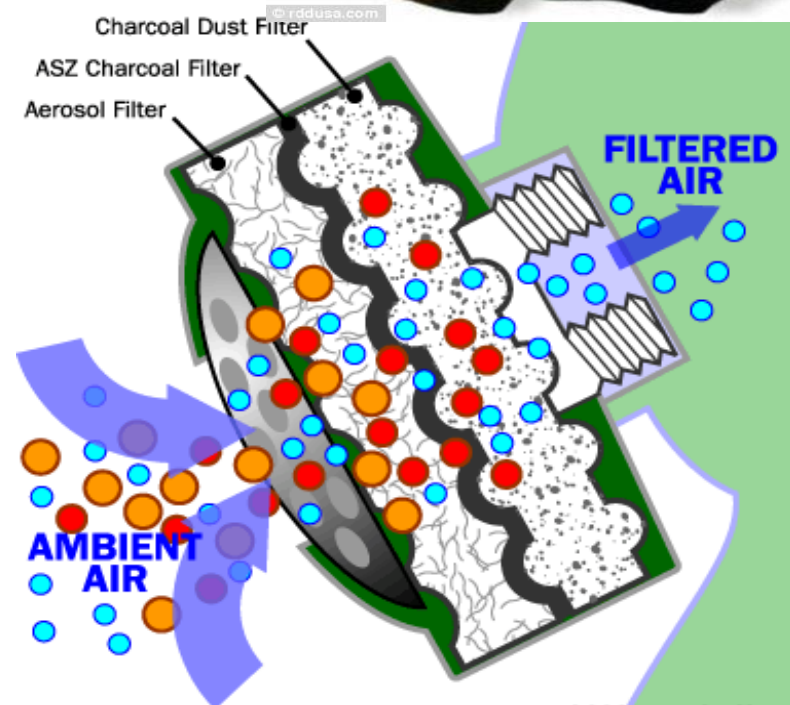
- ▶ Activated charcoal used as antidote for toxins
- ▶ Binding of the toxin to prevent stomach and intestinal absorption. Binding is reversible so a cathartic such as sorbitol may be added as well.
- ▶ Tablets or capsules of activated charcoal are used in many countries as an over-the-counter OTC drug to treat diarrhea and flatulence.
- ▶ Kaolin mainly used as antidiarrhea





General applications of adsorption

- ▶ In glass masks:
- ▶ All gas masks are devices containing suitable adsorbent so that the poisonous gases
- ▶ present in the atmosphere are preferentially adsorbed and the air for breathing is
- ▶ purified. such as methane (CH_4).
- ▶ Activated charcoal filters become less effective with use and have to be recharged or replaced.



Interesting videos will help you in understanding the topic well

<https://www.youtube.com/watch?v=Az1h5qMfFQM>

<https://www.youtube.com/watch?v=djlzXvwIz5U>

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THANK YOU